



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to G.V.K. FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

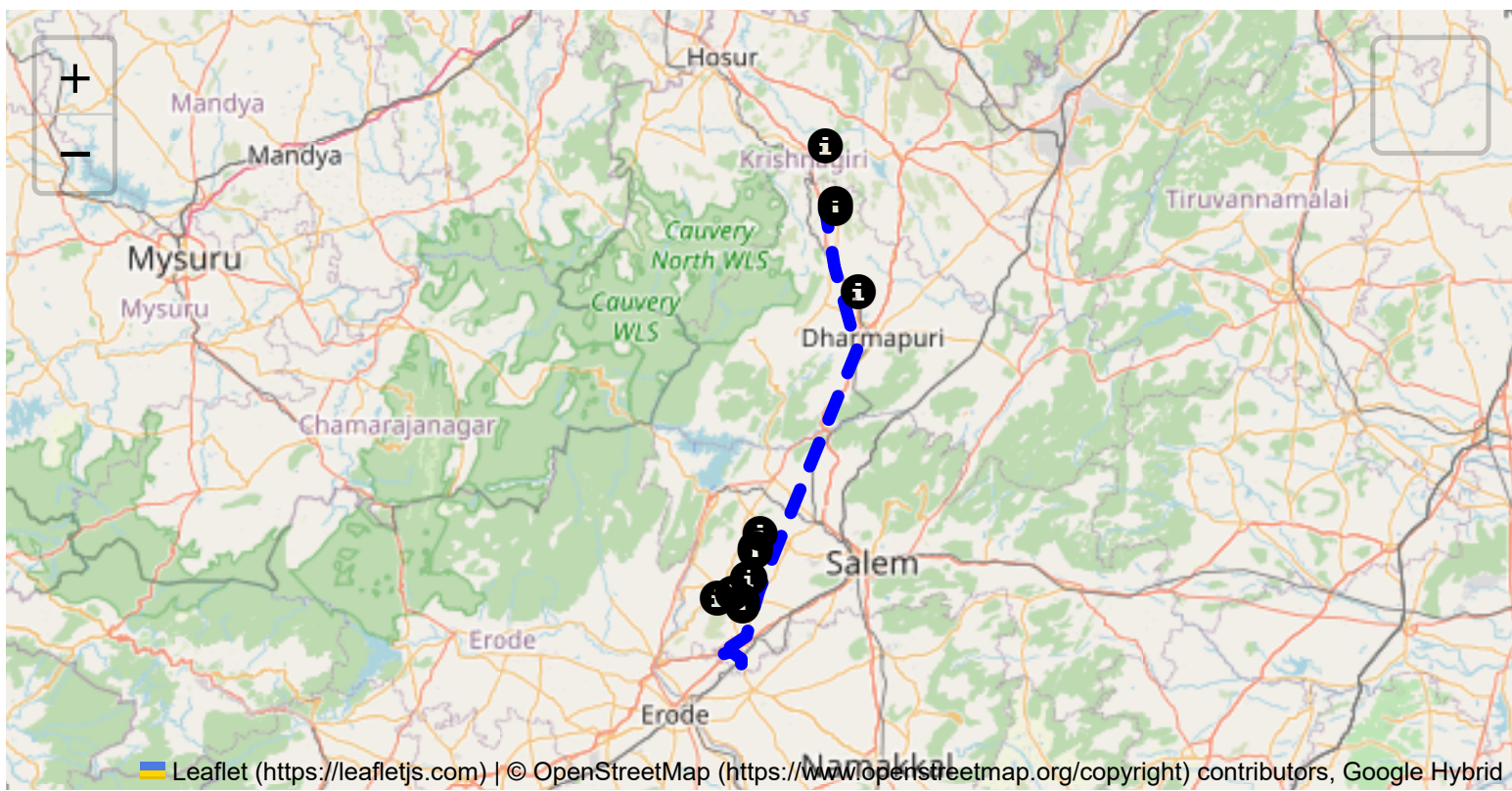
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

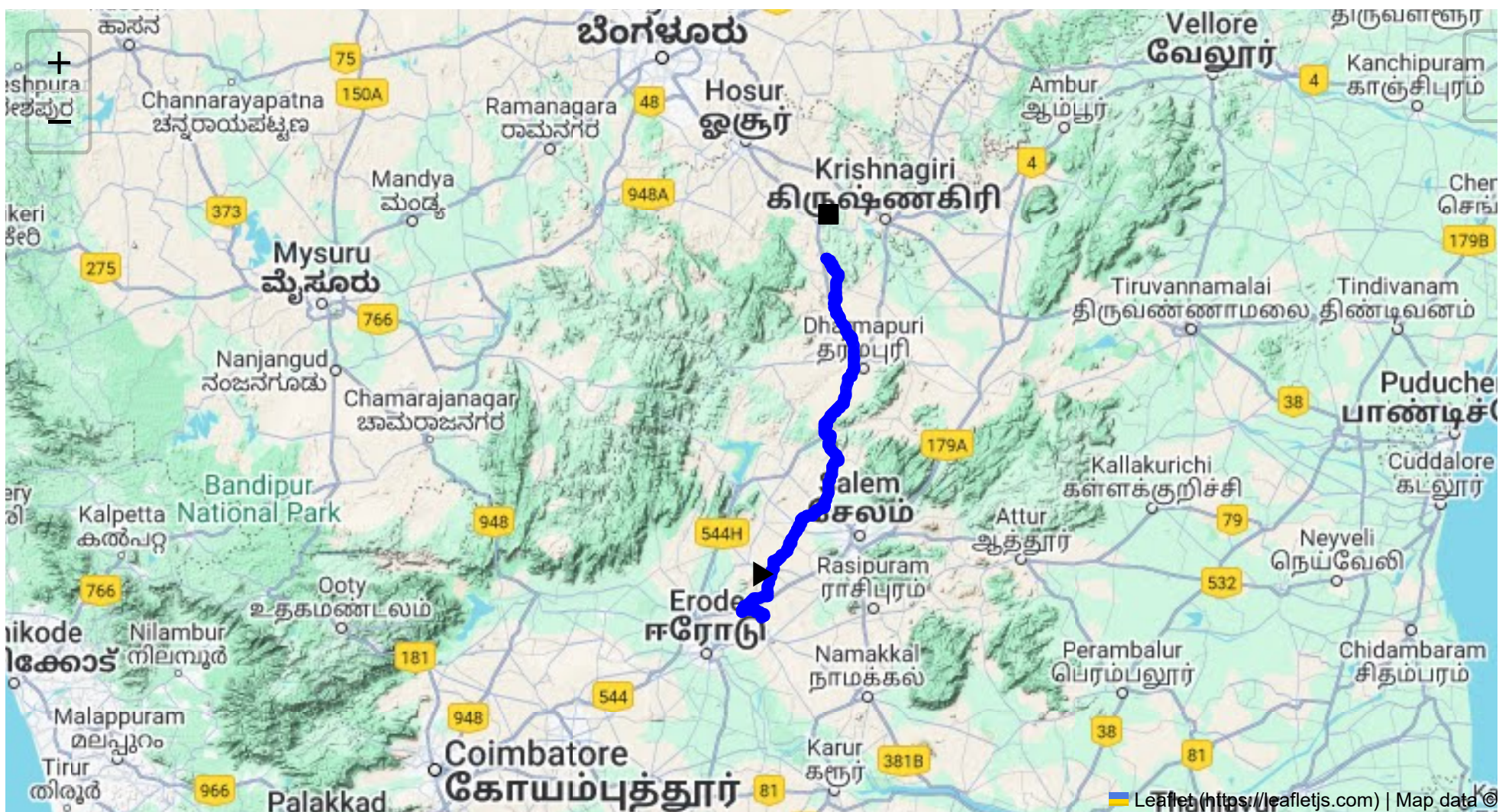
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 139.95 km
Estimated Duration: 2.9 hours
Adjusted Duration (Heavy Vehicle): 3.7 hours
Start: (11.4381, 77.8734)
End: (12.41883, 78.05002)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to C29X+PW3, Jittandahalli covers approximately 140 kilometers, predominantly traversing the SH 86 and Bhavani Main Road. It passes through several small towns and rural areas with a mix of state highways and local roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate with a hot and dry season from March to June, and a monsoon season from July to September. Heavy rains during the monsoon can lead to waterlogged roads and reduced visibility. During the dry season, heat and glare might pose challenges.

3. Analysis of Traffic Patterns

Traffic is generally moderate but can get congested around the towns of Sankari and Palacode particularly during peak hours (8-10 AM, 5-7 PM). Markets and local events might cause unexpected congestion.

4. Assessment of Road Quality and Infrastructure

Sections of SH 86 are well-maintained, though certain stretches might have potholes and uneven surfaces. Local roads in rural areas may not be in as good condition, especially after heavy rains.

5. Suggestions for Alternative Routes for Emergencies

Alternative routes via NH 44 can be considered in case of blockages but would lengthen the journey. Local shortcuts might not be suitable for heavy vehicles.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu requires permits for the transport of hazardous materials, specifying approved routes and times. Typically, movement during night hours might be restricted, and vehicles must clearly display hazard placards.

7. Overview of Historical Incidents

While there is generally low incidence of major accidents involving hazardous materials, the areas around Sankari and Palacode have reported occasional vehicular accidents due to road conditions and human error.

8. Environmental Considerations and Sensitive Areas

The route passes primarily through rural and agricultural zones. Care must be taken near water bodies to prevent contamination from hazardous material spills.

9. Analysis of Communication Coverage

Most of the route has adequate mobile network coverage, though rural stretches might have occasional dead zones, especially in hilly areas around the Erumaipatti section.

10. Estimated Emergency Response Times

Urban areas like Palacode may have faster response times (20–30 minutes) from emergency services due to better infrastructure. In rural areas, response can be delayed up to 45 minutes to an hour.

11. Overall Summary of Risk Assessment

This route is relatively safe under normal conditions with low to moderate traffic. Issues such as road quality, weather conditions, and adherence to regulations are critical for safely transporting hazardous materials. Pre-planning for emergencies and familiarizing drivers with alternative routes and communication protocols can significantly mitigate risks. Regular vehicle maintenance checks are recommended before embarking due to variable road conditions across the route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	High	11.45571, 77.81702	15 KM/Hr
7	Turn	High	11.45615, 77.81677	15 KM/Hr
8	Turn	Medium	11.46916, 77.82431	30 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.59214, 77.90656	30 KM/Hr
13	Turn	Medium	11.63833, 77.94818	30 KM/Hr
14	Turn	Medium	11.64171, 77.95184	30 KM/Hr
15	Turn	Medium	11.64243, 77.95198	30 KM/Hr
16	Turn	Medium	11.64721, 77.95118	30 KM/Hr
17	Turn	Medium	11.67870, 77.96846	30 KM/Hr
18	Turn	High	11.70227, 77.97864	15 KM/Hr
19	Turn	Medium	11.71715, 77.99969	30 KM/Hr
20	Turn	Medium	11.71921, 78.01133	30 KM/Hr
21	Turn	Medium	11.72737, 78.02248	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
22	Turn	Medium	11.73775, 78.03465	30 KM/Hr
23	Turn	High	11.73651, 78.03878	15 KM/Hr
24	Turn	Medium	11.74189, 78.04660	30 KM/Hr
25	Turn	Medium	11.74309, 78.04725	30 KM/Hr
26	Turn	Medium	11.77961, 78.05775	30 KM/Hr
27	Turn	Medium	11.86873, 78.08374	30 KM/Hr
28	Turn	Medium	11.87966, 78.08187	30 KM/Hr
29	Turn	Medium	11.92609, 78.06513	30 KM/Hr
30	Turn	Medium	11.93719, 78.05372	30 KM/Hr
31	Turn	Medium	11.95457, 78.04685	30 KM/Hr
32	Turn	Medium	11.95688, 78.04803	30 KM/Hr
33	Turn	High	11.95828, 78.04983	15 KM/Hr
34	Turn	Medium	11.95997, 78.04946	30 KM/Hr
35	Turn	Medium	11.96335, 78.05141	30 KM/Hr
36	Turn	High	11.96408, 78.05131	15 KM/Hr
37	Turn	Medium	11.96511, 78.04966	30 KM/Hr
38	Turn	High	11.96608, 78.04919	15 KM/Hr
39	Turn	Medium	11.96819, 78.05088	30 KM/Hr
40	Turn	Medium	12.26532, 78.08427	30 KM/Hr
41	Turn	Medium	12.27200, 78.08089	30 KM/Hr
42	Turn	Medium	12.29559, 78.07373	30 KM/Hr
43	Turn	High	12.29603, 78.07395	15 KM/Hr
44	Turn	Medium	12.29641, 78.07527	30 KM/Hr
45	Turn	Medium	12.31165, 78.07858	30 KM/Hr
46	Turn	Medium	12.31237, 78.07798	30 KM/Hr
47	Turn	High	12.31355, 78.07485	15 KM/Hr
48	Turn	Medium	12.31427, 78.07473	30 KM/Hr
49	Turn	Medium	12.37219, 78.08515	30 KM/Hr
50	Turn	Medium	12.37298, 78.08503	30 KM/Hr
51	Turn	High	12.37568, 78.08362	15 KM/Hr
52	Turn	High	12.37813, 78.08521	15 KM/Hr
53	Turn	Medium	12.37943, 78.08450	30 KM/Hr
54	Turn	Medium	12.38645, 78.07531	30 KM/Hr
55	Turn	Medium	12.39153, 78.07346	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
56	Turn	Medium	12.39347, 78.07169	30 KM/Hr
57	Turn	Medium	12.40605, 78.06717	30 KM/Hr
58	Turn	Medium	12.41316, 78.06064	30 KM/Hr
59	Turn	Medium	12.41690, 78.05592	30 KM/Hr
60	Turn	Medium	12.41787, 78.05223	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
14	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
15	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
16	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
18	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
19	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
24	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
26	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
27	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
28	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Government Hospital Palacode	12.2964824, 78.0761735	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
11	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
23	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
25	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
29	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45615, 77.81677



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59214, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63833, 77.94818



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64171, 77.95184



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64243, 77.95198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67870, 77.96846



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71715, 77.99969



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71921, 78.01133



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72737, 78.02248



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73775, 78.03465



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03878



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74189, 78.04660



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74309, 78.04725



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77961, 78.05775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86873, 78.08374



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87966, 78.08187



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92609, 78.06513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93719, 78.05372



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95457, 78.04685

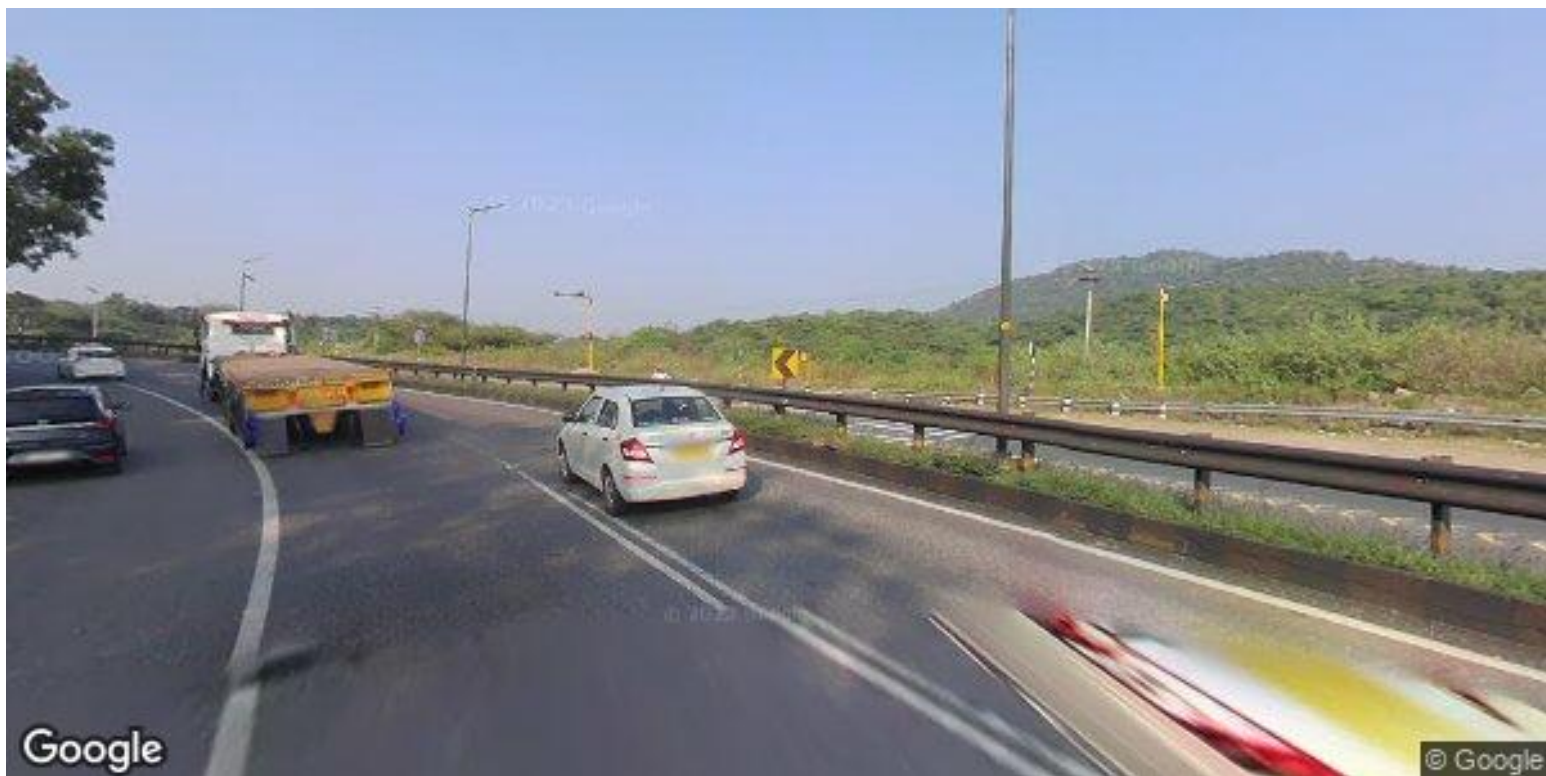


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95688, 78.04803



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95828, 78.04983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95997, 78.04946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96335, 78.05141



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96408, 78.05131



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96511, 78.04966



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96608, 78.04919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96819, 78.05088



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26532, 78.08427



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.27200, 78.08089



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.29559, 78.07373



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.29603, 78.07395



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29641, 78.07527



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31165, 78.07858



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31237, 78.07798



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.31355, 78.07485



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.31427, 78.07473



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.37219, 78.08515



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.37298, 78.08503



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.37568, 78.08362



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.37813, 78.08521



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.37943, 78.08450



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.38645, 78.07531



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39153, 78.07346



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39347, 78.07169



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40605, 78.06717



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.41316, 78.06064



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.41690, 78.05592



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.41787, 78.05223
